

Project Proposal

Intelligent Sensor System for Early Prediction of Sepsis in Neonates and High-Risk Adults

I. Background and Significance

Sepsis is a life-threatening condition caused by the body's dysregulated response to infection. Globally, it is estimated to affect around 49 million people and cause around 11 million deaths every year, thus accounting for nearly 20% of global deaths. It is a major cause of mortality in neonates and high-risk adults, particularly in intensive care settings. Early detection is crucial, as timely intervention can significantly reduce organ failure, prolonged hospital stays, and mortality rates.

Current clinical diagnostic methods rely on biomarker-based approaches, detecting host immune / inflammatory responses such as Interleukin-6 (IL-6) and Procalcitonin (PCT). However, these methods are indirect, detecting sepsis only after the body has already reacted to the infection. Additionally, traditional blood cultures take several hours to days to identify the causative pathogen, delaying treatment.

This project aims to develop a rapid, nanotechnology-enabled sensor system for early sepsis detection by directly identifying bacterial markers instead of host response biomarkers. The core innovation lies in utilizing Surface-Enhanced Raman Spectroscopy (SERS) to detect bacteria-specific protein markers (e.g., lipopolysaccharides (LPS), peptidoglycans) and integrating AI algorithms to analyze Raman spectral data.

II. Objectives

The primary goal of this project is to develop an AI-powered intelligent sensor system for real-time, rapid pathogen detection in neonates and high-risk adults.

The specific objectives are:

- a. **Biomarker Discovery:** Identify bacterial biomarkers (e.g., LPS, peptidoglycan) specific to sepsis causing Gram-positive and Gram-negative bacteria, through Raman spectroscopy using pure, commercially sourced bacterial components as reference standards.
- b. **SERS-Based Sensing Technology:** Develop a SERS-enabled Nanosensor chip optimized for detection of the discovered biomarkers.
- c. **AI-Integrated Raman Analysis:** Develop AI and neural network (NL) algorithms for automated interpretation of Raman spectral signatures, improving specificity and accuracy.
- d. **Clinical Validation:** Evaluate the SERS-based sensor on clinical samples from neonates and high-risk adults to assess accuracy, sensitivity, and specificity.
- e. **Prototype Development:** Design and fabricate a point-of-care (PoC) device for real-time sepsis detection in clinical settings (ICU, neonatal wards).

III. Project Methodology & Phases

Phase 1: Biomarker Identification through Pure Standards

- a. Selection of bacterial biomarkers: Identify key structural components of sepsis-causing bacteria, focusing on:
 - Gram-negative bacteria: Lipopolysaccharides (LPS), outer membrane proteins
 - Gram-positive bacteria: Peptidoglycan, teichoic acids, surface proteins
- b. Procurement and Raman Spectroscopy Studies:
Use commercially available purified LPS, peptidoglycans, and bacterial proteins to obtain high-resolution Raman spectra.
Optimize SERS substrates to enhance signal specificity for bacterial biomarkers.
- c. Establishment of Raman Spectral Library:
Create a reference database of Raman fingerprints of sepsis-related bacterial markers.

Phase 2: Validation on Cultured Clinical Isolates

- a. Collection of clinically isolated sepsis-causing bacteria from hospital microbiology labs.
- b. Perform Raman spectral comparison of pure biomarkers vs. whole bacteria to confirm signal reproducibility.
- c. Optimize functionalization of SERS chips for bacterial marker detection.

Phase 3: Clinical Validation Studies

- a. Patient Sample Collection: Collect blood, saliva, or sweat samples from neonates and high-risk adults diagnosed with sepsis.
- b. Process samples using mild lysis or separation techniques to enrich bacterial components.
- c. Raman Spectroscopy Analysis: Compare clinical sample spectra with spectral library data to detect sepsis-specific bacterial signatures.
- d. Train AI models to differentiate bacterial signals from host-derived interference.

Phase 4: AI-Enabled SERS Biosensor Development

- a. Develop a machine learning algorithm to analyze and classify Raman spectra for rapid sepsis detection.
- b. Integrate AI-powered SERS biosensor with real-time signal processing to improve specificity.
- c. Optimize sensor performance for sensitivity, reproducibility, and real-time data processing.

Phase 5: Prototype Development of a Point-of-Care (PoC) Device

- a. Design a portable Raman spectroscopy-based diagnostic system with AI integration.
- b. Conduct pilot studies in ICU / hospital settings to test usability and efficiency.
- c. Develop a low-cost, rapid screening tool suitable for bedside testing in neonates and critically ill patients.

IV. Deliverables

- a. Structured Raman spectral library of sepsis-causing Gram-positive and Gram-negative bacterial biomarkers.
- b. SERS chip optimized for detecting bacterial biomarkers from clinical isolates and patient samples.
- c. AI-based algorithm for analyzing Raman spectra, enabling rapid bacterial identification.
- d. Prototype demonstration suitable for point-of-care diagnostic device for hospital and ICU use.

V. Impact

- a. Shifts focus from host-response biomarkers to direct bacterial detection for improved specificity.
- b. Reduces sepsis detection time from hours/days to minutes.
- c. Provides a non-invasive, frugal, and point-of-care solution suitable for neonates and high-risk patients.
- d. Combines nanotechnology and AI to create a next-generation diagnostic platform for sepsis prediction.

VI. Budget and manpower details:

Overall Budget	1 st year	2 nd year	3 rd year	Total
Manpower	10,21,600.00	10,21,600.00	10,92,600.00	31,35,800
Hardware, Software & Equipment	30,60,000.00	0.00	0.00	30,60,000
Others	9,00,000.00	10,50,000.00	9,00,000.00	28,50,000
Total	49,81,600.00	20,71,600.00	19,92,600.00	90,45,800
Institute overhead (20%)	9,96,320.00	4,14,320.00	3,98,520.00	18,09,160
Total with overhead	59,77,920.00	24,85,920.00	23,91,120.00	108,54,960

Grand total = 108,54,960.00

Manpower details:

	Year 1	Year 2	Year 3
Current Team* 1. Prof Ajay Agarwal, PI, IITJ 2. Prof (Dr) Neeraj, Co-PI, AIIMS 3. Dr Paramita Kar Choudhury, Co-PI, TCS 4. Dr Sujit Shinde, Co-PI, TCS	As per TCS	As per TCS	As per TCS

5. Dr Sanjay Kimbahune, Co-PI, TCS 6. Dr Sudipta Bhattacharyya, Co-PI, IITJ			
New Requirement**			
1. Project Assistant/ Technical Assistant ¹ : 1 No	3,88,800	3,88,800	3,88,800
2. JRF/ SRF ² : 1 No			
3. Interns (BTech, MTech and/ or PhD students), (as and when required)	5,32,800 1,00,000	5,32,800 1,00,000	6,04,800 1,00,000
Total Headcount	10,21,600	10,21,600	10,92,600

¹ @Rs 32,400/ month for Project Assistant/ Technical Assistant (including 20% HRA).

² @Rs 44,400/ month for JRF or Rs 50,400/ month for SRF (including 20% HRA).

Note: Unutilized amount of Project/ Technical Assistant or JRF/ SRF will be used for hiring additional Interns.

* <Names of current headcount>

**Skills for new requirement= Engineering, Bio-medical or similar.

Hardware, Software and Equipment

	Year 1	Year 2	Year 3
Hardware	23,00,000	00	00
Software	4,60,000	00	00
Equipment	3,00,000	00	00
Total Cost in INR	30,60,000	00	00

Hardware: Workstation(s) for simulation and data processing, handheld Raman spectrometer for point-of-care solution = **Rs 23.0 Lakh**

Software:

Gaussian Version 16 (64Bit) Academic Site License with GaussView Version 6 = **Rs 4.6 Lakh**

Equipment: Wet etch setup = **Rs 3.0 Lakh**

Others	Year 1	Year 2	Year 3
Collaboration[§]	1,00,000	50,000	
Travel[#]	2,00,000	2,50,000	3,00,000
Misc[*]	6,00,000	7,50,000	6,00,000
Total Cost in INR	9,00,000	10,50,000	9,00,000

[§]**Collaboration:** for collaborator at AIIMS Jodhpur for miscellaneous requirements like laptop, travel, etc.

[#]**Travel:** For commutation between IIT and AIIMS; for interaction meetings; information sharing meetings, conference, etc.

^{*}**Misc.** expenditure includes:

- Chemicals, lab wares, components, etc.
- Substrates (silicon wafers, glass plates, polymers, etc.),
- Metal coating materials, targets, etc.
- Biomarkers,

- Equipment usage cost for device fabrication and characterizations.
- Contingency requirements.